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Equitable machine learning counteracts ancestral bias in precision medicine

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Gold standard genomic datasets severely under-represent non-European populations, leading to inequities and a limited understanding of human disease. Therapeutics and outcomes remain hidden because we lack insights that could be gained from analyzing ancestrally diverse genomic data. To address this significant gap, we present PhyloFrame, a machine learning method for equitable genomic precision medicine. PhyloFrame corrects for ancestral bias by integrating functional interaction networks and population genomics data with transcriptomic training data. Application of PhyloFrame to breast, thyroid, and uterine cancers shows marked improvements in predictive power across all ancestries, less model overfitting, and a higher likelihood of identifying known cancer-related genes. Validation in fourteen ancestrally diverse datasets demonstrates that PhyloFrame is better able to adjust for ancestry bias across all populations. The ability to provide accurate predictions for underrepresented groups, in particular, is substantially increased. Analysis of performance in the most diverse continental ancestry group, African, illustrates how phylogenetic distance from training data negatively impacts model performance, as well as PhyloFrame's capacity to mitigate these effects. These results demonstrate how equitable artificial intelligence (AI) approaches can mitigate ancestral bias in training data and contribute to equitable representation in medical research.

Initiatives such as The Cancer Genome Atlas (TCGA)[1–4], All Of Us[5], and Therapeutically Applicable Research to Generate Effective Treatments (TARGET)[6] have generated a wealth of genomic resources for disease analysis, bringing about the field of precision medicine and revolutionizing cancer treatment[7]. However, these databases do not equitably represent the diverse human ancestries that comprise the global population[8]. Individuals with European ancestry are dramatically overrepresented in omics databases (Supplemental Figure 1B). TCGA cancers have a median of 83% European ancestry individuals (range 49-100%)[1], and the GWAS Catalog[9], the largest genomic database that explicitly defines ancestry, is currently 95% European. A recent analysis of cell line data estimated that only 5% of existing transcriptomic data[10] and 2% of analyzed genomes derive from individuals of African descent[11], despite Africa being home to greater human genetic diversity than all other continents combined[12]. Numerous recent studies have observed substantial disparities in precision medicine effectiveness across ethnic groups[13–19], demonstrating a fundamental gap and inequity in some of the most cutting-edge approaches to precision medicine for cancer and other genetic disorders. A recent study in GWAS data demonstrated that model efficacy is correlated with population sample size; populations with little or no representation in the data have larger disparities in the disease model performance and garner little benefit from the benchmark disease model[20]. As such, ascertainment bias caused by unrepresentative sampling of ancestries

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is an acute unsolved challenge in major spheres of human disease genomics, including GWAS[21–23] and transcription models[24–26].

While big data genomic resources have revolutionized the study of cancer and other diseases, the systemic imbalance in cancer genomic data collection across human populations is severe. Given that human ancestry has a substantial impact on gene expression in healthy and disease tissues[27–31], it is essential that we consider population-specific variation when modeling disease in order to improve outcomes for all populations. Recognition of differences in disease occurrence between ancestries has led to some differential prioritization of treatments[32–34]. However, these race-based approaches are proxies of genetic ancestry-related disease risk factors, limiting their potential efficacy. Moreover, the efficacy of such approaches should be expected to decrease, as the proportion of individuals with admixed ancestry (ancestry from multiple populations) is increasing globally[35], and so there is a critical need for precision medicine models that perform well in admixed populations. Predicted disease risk in admixed individuals, who have historically been excluded from studies due to their population stratification, changes depending on the proportions of continental ancestry and alleles an individual inherits. These individuals cannot be sufficiently represented by single-ancestry medical models. For these and all other individuals, access to diverse data is critical for creating robust precision medicine and answering scientific questions in the field[13].

Despite overwhelming genomic similarity, human populations differ in ways that can substantially impact cancer treatment. African populations have the greatest genetic diversity[12], while non-African populations' genetic diversity is negatively correlated with distance from Africa due to population bottlenecks during the migration out of Africa[36]. In addition, as humans migrated throughout Africa and to other areas of the world (e.g., Asia, Europe, and the Americas), populations underwent selection for traits beneficial in their new environments[37–39]. Finally, because admixture between neighboring populations is a near constant in human history, a given individual in one population is likely to possess some alleles that are under-represented in their population compared to others. For example, any given European individual is likely to have some alleles that are more common in Africa than Europe. Even with diverse large-scale genomics data, individual studies will not have such ancestral diversity, and single ancestry models cannot capture admixed individuals. Including known human diversity information is another source of data for models to use in learning disease and other outcomes from predictive modeling, and this trove of information should not be ignored. Additionally, while diverse data is likely to reduce the impacts of sequencing disparities, the different sizes of human ancestral populations and the variability in diversity within them makes it unlikely that any studies will ever be truly equitable with regard to ancestry. To discern ancestral differences in disease presentation and susceptibility, we must first develop a better understanding of the genetic variation that exists among ancestries[40]. Addressing ancestry imbalance in genomic data to improve model performance will require years of dedicated large-scale sequencing efforts. Equitable machine learning approaches can help bridge the gap.

In this work, we address this need with an ancestry-aware equitable machine learning framework for transcriptomics, PhyloFrame. PhyloFrame is an equitable machine learning method that creates signatures of disease based on population genomics data and disease-relevant training data, resulting in ancestry-aware signatures that generalize to all populations, even those not represented in the training data. It does this without needing to call ancestry on the training data samples. We demonstrate across three TCGA cancers with the greatest ancestral diversity (breast (BRCA), thyroid (THCA), and uterine (UCEC)) that PhyloFrame provides improved performance compared to a comparable benchmark model, even in the likely scenario where few ancestries are represented in the training data. For each cancer, we selected a different prediction task, aiming for a variety of ancestral bias in outcomes and in difficulty of the prediction task. In BRCA, we predict basal versus luminal subtype. This is an easy prediction task in a common cancer which has been extensively documented to be ancestry biased[27,41,42]. For UCEC we also predict a subtype (endometrioid vs serous), a harder task given the rarity of one of the subtypes and, given the many molecular characteristics with breast and ovarian cancers, likely also ancestry biased. The third prediction task we selected, whether or not THCA tumors will later undergo metastasis given bulk RNAseq data from a primary tumor, is significantly more difficult and ancestry biases more complex, as racial disparities in metastatic progression are highly variable across cancer types[43]. In all three cancers, we show that PhyloFrame mitigates the effects of ancestry imbalances in the training data from the individual disease studies, and improves outcomes for all ancestries. This approach offers an immediately actionable alternative to the mass sequencing that would be required to capture disease-driving genes across underrepresented populations in every cancer. Additionally, PhyloFrame's inclusion of population genomics data, and therefore the population structure information that is inherent to human diversity[44], provides an essential and durable advantage over global bulk analyses regardless of the number of samples involved, allowing PhyloFrame to identify population-specific variation more readily. In summary, our equitable machine learning framework, PhyloFrame, provides an alternative approach for the inclusion of human ancestral population diversity in medical studies, and we demonstrate the viability of this approach.

Results

Ancestry-specific signatures share pathway-level dysregulation

We sought first to identify the extent to which ancestry impacts the transcriptomic signatures of disease when using unbalanced data. To distinguish ancestry-specific and disease-specific signatures, we compare differences between model signatures when training on data from different ancestries. We selected the two largest TCGA BRCA ancestral populations, European (EUR; 665 samples, 107 basal and 558 luminal) and African (AFR; 90 samples, 37 basal and 53 luminal), and trained an elastic network to predict basal versus luminal breast cancer subtypes. Two models were trained, one using the EUR data and one using the AFR data. Supplemental Figure 2 shows the two resultant signatures of disease (full signatures in Supplemental Table 1) projected onto the HumanBase mammary epithelium functional interaction network. While there is little direct overlap in the signatures, projecting them onto the functional interaction network highlights how the signatures are within shared subnetworks. This suggests that the ancestry-specific disease signatures are strongly interconnected, and that it is possible to create equitable signatures of disease without requiring ancestry information for the training data. It also potentially negates the need for sufficiently large training datasets from each human ancestry population across the globe, which would be necessary to train ancestry-specific models. Instead, it suggests that equitable signatures of disease can be detected from models trained on unbalanced ancestral training data if population genomics data are used to help adjust for distribution shifts in the data[45–47]. This avoids reliance on single-ancestry models, which assume that individuals come from a single ancestry group; unrealistic in an increasingly diverse world.

Identifying ancestry-specific genetic variants

To leverage the functional interaction networks and find ancestry-agnostic signatures, it is critical to first identify which network nodes adjacent to disease signature genes are ancestry-enriched. We expect that genes with high variance among ancestries are more likely to underlie ancestry specific variation, and that some important disease causing functions will be under-represented in existing European biased databases. To identify these loci, we define Enhanced

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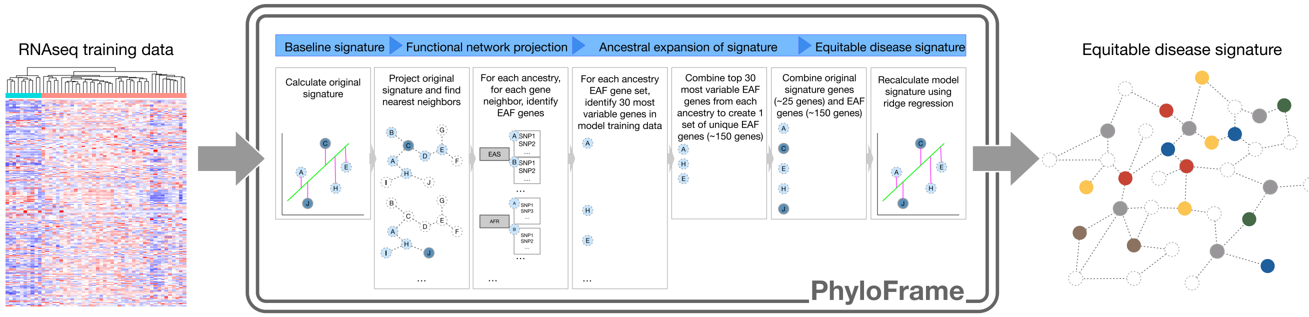

Fig. 1 | A visual representation of the PhyloFrame method. A user provides expression data and outcome labels, and with this information PhyloFrame generates an equitable signature predictive of those outcomes. PhyloFrame generates an initial signature based on training data only using a logistic regression model. Next, PhyloFrame prunes a functional interaction network based on an PhyloFrame-defined edge weight (0.2 - 0.5), which the user has the option to adjust. The signature obtained from the initial logistic regression model is then projected onto the pruned network, and each signature gene is used as a start node for network propagation. This propagation is used to identify outcome-relevant genes that are more variable across ancestral populations. For each original signature gene, up to two nearest neighbors are selected and included in further filtering. Ancestry-specific filtering is performed on these gene neighbors. For each ancestry, for each gene, SNPs are filtered by their EAF. A gene must have a SNP that is uniquely enhanced in one ancestry in order to be considered enhanced by PhyloFrame. This creates a larger set of genes, for each ancestry group, that interact with those in the original signature and are enriched in that ancestry. For each of these sets of genes (one per ancestry group), the 30 most variable genes from the training data are identified and compiled to create a single set of unique EAF genes with equal contributions from all ancestry populations. This results in approximately 150 genes. This gene set is then combined with the original gene set (the set initially used as start nodes) and the logistic regression model is rerun using ridge regression to force inclusion of ancestry genes.

Allele Frequency (EAF), a statistic to identify population-specific enriched variants relative to other human populations (see Methods; Equation (1)). EAF captures population-specific allelic enrichment in healthy tissue. As such, higher EAF means that individuals from a population are more likely to have a variant than individuals from all other ancestries. Because EAF is calculated from healthy tissue, this approach can integrate information from under-represented populations that are not present in a smaller disease-specific databases, including TCGA. To confirm this, we compared EAF in COSMIC cancer-related genes to non-COSMIC genes and did not find an enrichment in EAF (Supplemental Figure 3; two-sided unpaired t-test of EAF in COSMIC vs non-COSMIC genes(df = 70, 379, 353, p = 1, effect size = $3.48e-18$, 95% CI = $-2.002e-6 - 2.002e-6$)). This is as expected, given EAF is calculated from healthy tissue. All populations in this study produce high EAF loci, however, more diverse and less well sampled populations (e.g. AFR, EAS, SAS) are associated with the largest number of EAF enriched genes (Supplemental Figure 4) and higher average EAF values. For example, average EAF across COSMIC genes (Supplementary Data 1) is greatest in the African ancestry group, consistent with the greater genetic diversity found in Africa compared to other continents[12]. This highlights the importance of broader sampling, as most of the standing variation in humans is derived from under-sampled populations.

Integrating population genomics and disease data

Given the potential uses of EAF and functional interaction networks to reduce ancestral bias in predictive modeling, we next sought to create a method that will use these information to create equitable signatures of disease and other disease outcomes. We present an equitable machine learning framework, PhyloFrame, which combines EAF with functional interaction networks to reduce ancestry effects and amplify predictive power when modeling outcomes using a user-provided data set (Fig. 1). PhyloFrame needs solely the gene expression data and class label (e.g. disease subtype) information for each sample, allowing for easy application to many diseases. It does not require ancestry information for the training data samples, nor does it assume that individuals are of a single ancestry. Not needing to calculate ancestry is a significant boon; not only does it remove a barrier to clinical adoption by eliminating a pipeline step that requires significant compute and expertise, but the methods are recent and constantly evolving and their accuracy and sensitivity is rapidly improving. Given the accuracy of ancestry calling methods in under-sampled populations and admixed individuals is unclear, not requiring ancestry information removes this potential source of bias. Furthermore, admixed individuals and populations (e.g. American) are under-served by models that assume individuals are of a single ancestry. Generating one signature that applies to all ancestries ensures that individuals will not be misclassified or need to be forced into a single ancestry group model when they have ancestry from multiple populations. By creating a single signature of outcomes that generalizes to all individuals, PhyloFrame avoids these issues. Finally, to facilitate easy adoption and application of the method, we release PhyloFrame as an R package (see Code Availability).

A visualization of the PhyloFrame method is shown in Fig. 1. PhyloFrame uses a logistic regression model with LASSO penalty to obtain an initial set of disease-relevant genes. We selected the logistic regression model as it and other models with heavy regularization are more robust to domain shifts[45–47], which often result in drastic performance drops when applied to out-of-distribution data, such as exists in human disease omics data[48]. Additionally, this highly interpretable method allows for in-depth analysis of the source of performance changes and uses an underlying approach (elastic network) that is frequently used and effective in RNAseq data applications. PhyloFrame then projects this original disease signature onto a functional interaction network, extending the network to include the first and second neighbors of each original signature gene. This new set of genes identified by the network extension is then filtered by EAF, which is used to rank the genes to determine a small set of ancestrally diverse genes that interact with those in the original model signature. These ancestrally diverse genes are ranked by variation in the training data. From each ancestry (see Methods for list of ancestry groups), a subset of genes with high EAF and gene expression variability in the training data are selected to be added to the PhyloFrame signature (see Methods). PhyloFrame then retrains the disease signature regression model and forces the inclusion of the equitable genes using a ridge regression penalty. The resultant signature includes disease-relevant genes that generalize to all populations. We evaluate against a benchmark model that uses the same machine learning approach but without population genomics data (visualized in Supplemental Figure 5). This benchmark model mimics the PhyloFrame model, generating a signature using a logistic regression with LASSO penalty then adding highly variable genes to the signature and forcing inclusion of those highly variable genes into the signature so that it matches the PhyloFrame signature size, making the models more directly comparable. It uses the exact same logistic regression implementation and signature size as PhyloFrame but without EAF amplification. Implementation details of PhyloFrame and the benchmark models are described in detail in the Methods section.

Evaluating model performance across many independent datasets

Benchmarking differences in performance across ancestral populations and between PhyloFrame and a benchmark model necessitates the analysis of datasets with ample ancestral diversity (diversity information for all TCGA cancers shown in Supplemental Figure 1A). Many datasets are not sufficiently large or ancestrally diverse for this. We selected three TCGA cancers (breast, thyroid, uterine) with enough data to be able to validate the models in and across multiple ancestries. Ancestry labels for all TCGA samples were obtained from Carrot-Zhang et al.[30]. The breast cancer (BRCA) models were trained to predict basal versus luminal subtypes, the thyroid (THCA) models whether a tumor would metastasize (M0 versus MX), and the uterine (UCEC) models endometrioid versus serous subtypes. In order to validate models in and across multiple ancestries, samples were divided into smaller independent, single ancestry datasets. To do this, samples from each cancer were initially divided by continental ancestry (European, EUR; African, AFR; East Asian, EAS; or Admixed, ADMIX). The continental ancestry group with the smallest number of samples determined the size of all the independent single-ancestry datasets in each cancer. This ensures that the smallest ancestry group would still be represented.

For each cancer, we divided the data into n independent datasets. All samples were assigned to one and only one dataset. This resulted in many independent datasets per disease. We created single ancestry datasets and datasets composed of admixed individuals, allowing for effective comparison of within-ancestry and across-ancestry model performance. This resulted in many models for some ancestral populations, providing a greater statistical basis for intra-population variation comparisons. Dataset sizes for each cancer are determined by the ancestry group with the fewest samples (see Methods). Splitting the data this way resulted in 21 BRCA datasets (17 EUR, 2 AFR, 1 EAS, 1 ADMIXED, 0 AMR, 0 SAS) of 43 ± 5 samples each from a total of 842 individuals. It resulted in 28 THCA datasets (23 EUR, 1 AFR, 3 EAS, 1 ADMIXED, 0 AMR, 0 SAS) of 16 ± 2 samples each from a total of 436 individuals, and 15 UCEC datasets (12 EUR, 2 AFR, 0 EAS, 1 ADMIXED, 0 AMR, 0 SAS) of 16 ± 2 samples each, from a total 491 UCEC individuals.

PhyloFrame and the benchmark were trained on each of these datasets, then applied to each other dataset separately. This approach resulted in 1,401 unique train-test dataset pairs with both PhyloFrame and the benchmark applied to each. Applying PhyloFrame and the benchmark to the test train pairs yields 2800 model-dataset combinations. We calculated Matthew's Correlation Coefficient (MCC), Area under the ROC curve (AUC), F-measure, Cohen's kappa, precision, recall, and accuracy for each trained model on each dataset (Supplementary Data 2). We used differences in these model performance metrics to quantify the effectiveness of the PhyloFrame models' ability to eliminate ancestry bias in unbalanced training data. See Methods for full implementation details.

Equitable machine learning has higher overall performance

To assess model predictive power, we compared the PhyloFrame and benchmark models across several performance metrics (Supplementary Data 2, Figure 2). When considering all three cancers together, PhyloFrame outperformed the benchmark across Matthew's Correlation Coefficient (MCC), Area under the ROC curve (AUC), F-measure, Cohen's kappa, recall, and accuracy (Bonferroni adjusted one-sided paired t-tests of: model MCC, PF vs BM ($df = 1215$, $p = 2.341e-20$, effect size = 0.274, 99% CI = 0.034 − *Inf*), model AUC, PF vs BM ($df = 1400$, $p = 0.003$, effect size = 0.09, 99% CI = 0.001 − *Inf*), model F-measure, PF vs BM ($df = 1397$, $p = 8.788e-28$, effect size = 0.302, 99% CI = 0.026 − *Inf*), model Cohen's kappa, PF vs BM ($df = 1400$, $p = 2.712e-31$, effect size = 0.322, 99% CI = 0.043 − *Inf*), model recall, PF vs BM ($df = 1400$, $p = 7.847e-23$, effect size = 0.271, 99% CI = 0.029 − *Inf*), model accuracy, PF vs BM ($df = 1400$, $p = 1.591e-12$, effect size = 0.195, 99% CI = 0.008 − *Inf*)). We also considered each cancer independently and found significant improvements in PhyloFrame performance in two of the cancers, BRCA and UCEC, and slight improvements in THCA (Fig. 2, Supplemental Figure 6). THCA, the most difficult prediction task, has low performance across both model types and although PhyloFrame metrics are slightly higher than the benchmark, the performance differences are not statistically significant. It's unclear if this is due to the nature of the predictive problem, the lack of differences between ancestries in relation to metastasis, or other reasons. Given the complexity of racial disparities in cancer metastasis rates[43], it's unlikely that there is a simple explanation for ancestry impacts on this prediction task. Performance differences are statistically significant in both UCEC and BRCA, both of which are easier prediction tasks and are known to have at least some ancestry bias. Confidence intervals for each comparison in the three cancers are shown in Fig. 2 and Supplementary Data 2. PhyloFrame model MCC in all cancers is higher than the benchmark (Bonferroni adjusted one-sided paired t-tests of: BRCA model MCC, PF vs BM ($df = 418$, $p = 9.143e-44$, effect size = 0.771, 99% CI = 0.085 − *Inf*), UCEC model MCC, PF vs BM ($df = 108$, $p = 6.827e-12$, effect size = 0.761, 99% CI = 0.078 − *Inf*), THCA model MCC, PF vs BM ($df = 687$, $p = 1$, effect size = 0.017, 99% CI = − 0.013 − *Inf*)), indicating that while both models are correctly predicting subtypes of positive class samples, PhyloFrame models are finding more of these individuals than the benchmark models and is overall better at balancing class positives and negatives. The PhyloFrame model advantages are largest in the BRCA prediction task, which has been extensively documented to be ancestry impacted. UCEC subtype definitions are also likely impacted by ancestry bias in data used to establish these subtypes, however both UCEC and BRCA subtypes appear in all ancestry groups and clinical management takes these subtypes into account. The PhyloFrame model is better at compensating for this than the benchmark. Trends across all ancestries show that PhyloFrame has better overall performance than the benchmark across most of these metrics in each cancer when considering the cancers separately.

Performance differences in analyses of the models trained on EUR data compared to models trained on non-EUR data show that models trained on EUR data have limited robustness when applied to diverse datasets. This trend is noticeable in all cancers; both PhyloFrame and benchmark models trained on EUR data tend to perform worse than those trained on non-EUR data (Bonferroni adjusted one-sided unpaired t-test of model MCC in all three cancers for PhyloFrame non-EUR vs EUR trained models ($df = 314.4$, $p = 2.306e-10$, effect size = 0.505, 99% CI = 0.145 − *Inf*), Bonferroni adjusted one-sided unpaired t-test of model MCC in all three cancers for benchmark non-EUR vs EUR trained models ($df = 286.49$, $p = 1.745e-12$, effect size = 0.591, 99% CI = 0.172 − *Inf*); mean MCC EUR PhyloFrame 0.33 and benchmark 0.279; mean MCC non-EUR PhyloFrame 0.548 and benchmark 0.522; Fig. 2, Supplementary Data 2). This is clinically significant because the EUR training sets most closely resemble the available omics data for the vast majority of cancers[8,13–19]. The largest difference in performance between EUR-trained and non-EUR-trained models is seen in the BRCA models, which are known to be heavily impacted by ancestry (Bonferroni adjusted one-sided unpaired t-test of model MCC in BRCA for PhyloFrame non-EUR vs EUR trained models ($df = 272.76$, $p = 1.515e-45$, effect size = 0.505, 99% CI = 0.129 − *Inf*), Bonferroni adjusted one-sided unpaired t-test of model MCC in BRCA for benchmark non-EUR vs EUR trained models ($df = 268.75$, $p = 1.063e-64$,

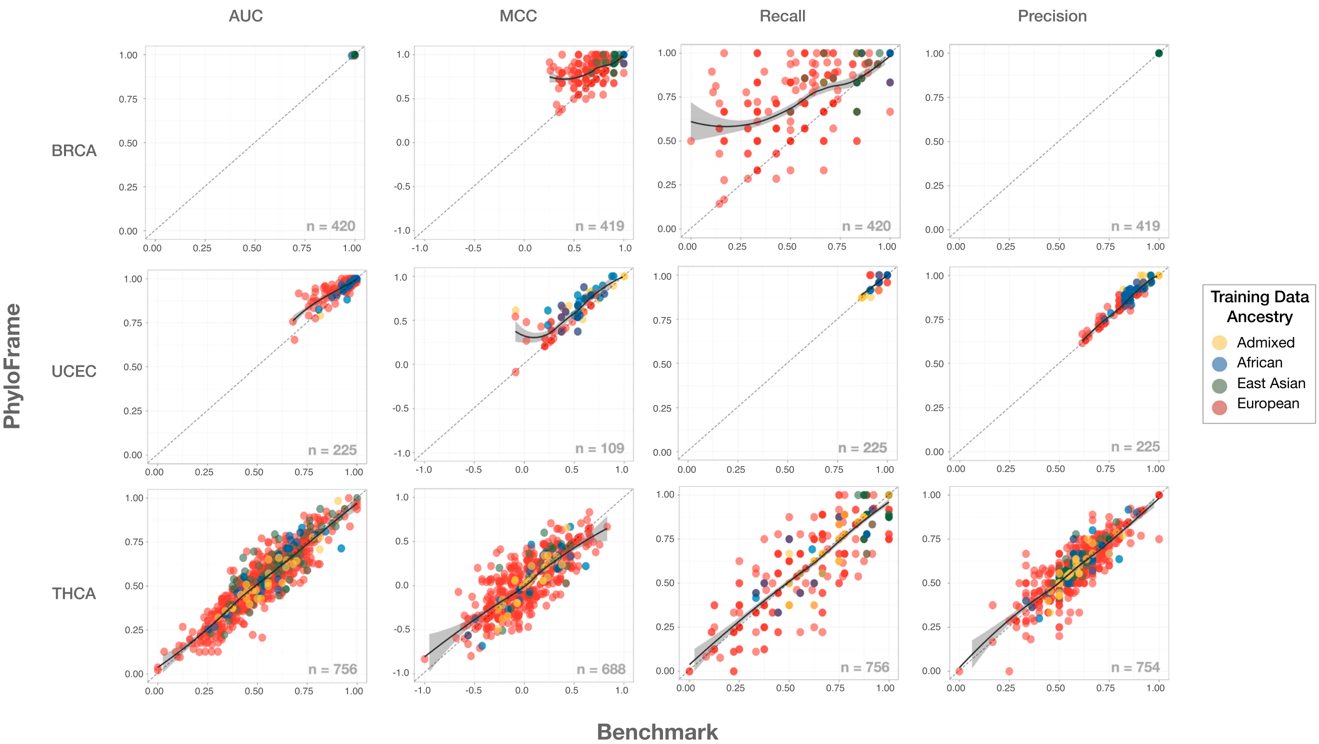

Fig. 2 | Performance metrics in each cancer. Scatterplots of AUC, MCC, recall, and precision of the PhyloFrame (y-axis) and benchmark (x-axis) models for the three TCGA cancer datasets. Plot points are colored by the ancestry of the data on which each model was trained. Each plot shows one performance metric of PhyloFrame versus benchmark for all models. Each trained model was applied to each other dataset separately (see Methods; 420 pairs of BRCA results, 756 THCA, and 225 UCEC). The number of model results shown in each plot is in the bottom right-hand corner. This number varies despite being trained on the same train-test dataset combinations because some results are NA due to class imbalance. NA values are not included in the plots. This occurred when either PhyloFrame or benchmark predicted all samples in the dataset as belonging to the majority class. Smoothing with local polynomial regression fitting (LOESS) method is used to fit a line to model results, shown as a solid black line in each plot for all BRCA plots (AUC, MCC, recall, precision), all THCA plots (AUC, MCC, recall, precision) and UCEC AUC, MCC, and precision plots. A 95% confidence interval is shown as a shaded bar surrounding each solid black line. Smoothing with a linear model is used to fit a line to model results for the UCEC recall plot because LOESS could not fit a line with the given data points. A 95% confidence interval is shown as a shaded bar surrounding the solid black line. Source data are provided as a Source Data file.

effect size = 2.21, 99% CI = 0.231 − *Inf*); mean MCC EUR PhyloFrame 0.82 and benchmark 0.699; mean MCC non-EUR PhyloFrame 0.97 and benchmark 0.958). PhyloFrame's performance advantage in EUR trained models is consistent with the hypothesis that the benchmark is overfitting, exacerbating existing biases toward Europeans (Supplemental Figure 6, Supplementary Data 2). This is particularly important because the EUR training datasets most closely mirrors ancestral diversity in existing data, impacting plausible research and clinical applications of machine learning prediction tasks. Additionally, PhyloFrame outperforming the benchmark in non-EUR datasets highlights the importance of equitable machine learning methods even in scenarios with relatively diverse data. Overall we observe a clear trend toward improved predictive performance in PhyloFrame relative to the benchmark across training sets and test data of all ancestries.

Considering the non-EUR trained models ancestry by ancestry in the two cancers with high model performance, BRCA and UCEC, PhyloFrame outperforms benchmark except when comparing the BRCA models trained on ADMIX and AFR data. This is notable because AFR and ADMIX individuals (most of which in the TCGA data have high levels of AFR ancestry) have both higher germ-line genetic diversity and much higher frequency of the overall less common basal subtype. Given that PhyloFrame significantly outperforms the benchmark in UCEC models trained on ADMIX and AFR individuals, we reject the hypothesis that germ-line genetic diversity is solely responsible for the non-significant BRCA results. Class label balance is a more plausible explanation for the non-significant BRCA performance differences, as BRCA AFR (41% basal) and BRCA ADMIX (33% basal) are the only BRCA or UCEC ancestry groups where the proportion of TCGA patients belonging to the less common class exceeds 20%. Unbalanced classes exacerbate data distribution shift issues[48,49], in part because the less common class is less likely to have samples that are representative of the full diversity of that class, and therefore we would expect more overfitting issues. Alternatively, the non-significant result may be a product of the overall high performance of all BRCA models, due to the easy prediction task. This leaves less room for improvement, and as a result potential PhyloFrame performance gains are smaller because benchmark models' performance is at or very near MCC=1. It may therefore be the case that both models would benefit from larger training data and more balanced classes, but as both PhyloFrame and the benchmark perform well on BRCA subtype prediction, statistical significance was not observed.

Equitable machine learning creates more stable disease signatures. Precision medicine models should be identifying biologically-relevant signatures of disease. A signature may be accurate on a set of training data, but it has limited utility if it does not generalize to other data or identify the biological drivers of disease. Inconsistency between the results of models trained on different groups of samples is a hallmark of model overfitting and leads to non-biologically informative results. We quantify the stability of our models by testing the overlap between disease signatures identified by each BRCA model (Fig. 3). Despite all of the BRCA models having high AUC (range 0.984 − 1.000), there is significantly less pairwise signature overlap within signatures identified by the benchmark models compared to

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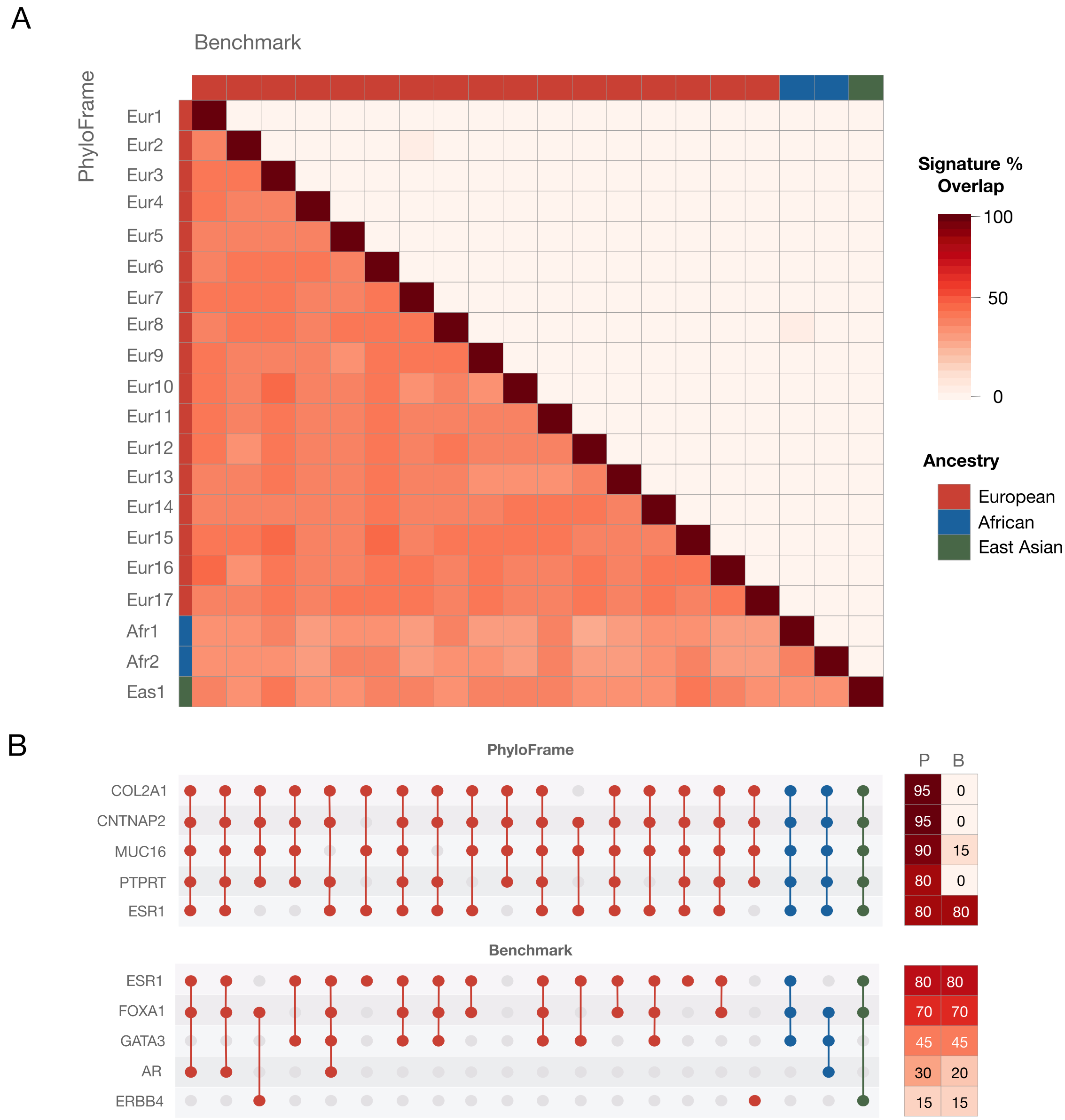

Fig. 3 | Equitable machine learning effectiveness. A Signature-signature overlap of BRCA model signatures when trained on each dataset. Sidebar annotations show ancestry of the training data. Top triangle shows signature-signature overlap between pairs of benchmark model signatures. Bottom triangle shows the same but for PhyloFrame. **B** Dotplot showing the 5 most frequently included COSMIC genes in the PhyloFrame (top) and benchmark (bottom) models, within each PhyloFrame and benchmark model signature. Full lists in Supplemental Figure 7. Heatmaps to the right of the dotplots indicate the percentage of PhyloFrame (P) and benchmark (B) models that include the given COSMIC gene. Source data are provided as a Source Data file.

PhyloFrame model signatures (Fig. 3A). PhyloFrame models trained on EUR data have, on average, 44.7% overlap with other PhyloFrame model signatures, whereas benchmark models have 6.1% overlap with other benchmark model signatures. The greater signature stability between training sets observed in PhyloFrame relative to the benchmark suggests that PhyloFrame is less impacted by overfitting to the training data.

PhyloFrame models also much more consistently identify known cancer-related genes. Each PhyloFrame BRCA model signature identifies more COSMIC cancer-related genes than its benchmark counterpart (9.7 COSMIC genes on average). PhyloFrame BRCA models are also more consistent in the COSMIC genes they identified, finding 30 unique COSMIC genes across all model PhyloFrame signatures (Fig. 3B, Supplemental Figure 7), compared to the 93 unique COSMIC genes identified across the benchmark BRCA models. As a result, 73% of COSMIC genes identified in PhyloFrame models are found in multiple signatures compared to only 24% of COSMIC genes identified in more than one benchmark signature. The greater consistency between PhyloFrame models is not attributable to high EAF genes as there is no EAF enrichment in COSMIC cancer–related genes versus others (Supplemental Figure 3). While the benchmark models most frequently identify canonical BRCA genes such as *ESR1* and *FOXA1*, those canonical BRCA genes are identified equally often by PhyloFrame (Fig. 3B). PhyloFrame also identifies other COSMIC genes at even higher rates than these genes, suggesting that they are deserving of further enquiry for their role in breast cancer, particularly in breast cancers in non-European populations. For example, *MUC16*, identified in 90% of PhyloFrame models and 15% of benchmark models, is known to be implicated in metastasis of triple negative breast cancers[50,51], which are substantially more common in non-European populations. COSMIC genes captured more frequently by PhyloFrame BRCA models (Supplemental Figure 7) are included in PhyloFrame because they have a high EAF in ancestries not found in the training data. For example, *MUC16* has high EAF in AFR and EAS ancestries.

Identifying model vulnerabilities via hard-to-predict samples

Model performance metrics such as AUC or MCC are important indicators of model success. However, AUC and MCC quantify overall model performance rather than performance variability within a set of samples. Given some samples are far harder to accurately predict than others, it is beneficial to assess sample-specific performance in addition to cohort-level metrics such as AUC and MCC. To assess this, we plot sample-specific accuracy (the percent of models that correctly predict outcome labels for each sample; Supplemental Figure 8), and identify which samples are often misclassified, if any. PhyloFrame and the benchmark both performed well on BRCA samples, although the benchmark models trained on EUR individuals under-performed for AFR and ADMIX ancestry individuals. UCEC exhibits greater error rates than BRCA but less than THCA. Models trained on EUR UCEC data show a decline in predictive success in ADMIX and AFR individuals compared to other ancestries. ADMIX individuals also are predicted less effectively by AFR trained models compared to other ancestries, highlighting the unique challenges associated with prediction for admixed individuals. Finally, THCA models have far more variability, as expected, given metastasis is a harder prediction task than tumor subtypes. Of note, this per-sample performance is not shared across models; There is a wide range of success for each set of sample predictions. This suggests that there are factors relevant to THCA metastasis that are not being identified by the models. While the UCEC and THCA models demonstrate the utility of PhyloFrame even in difficult data and prediction tasks, they also highlight the critical need for diverse data. Together, improved equitable AI models plus diverse data likely will result in fully equitable precision medicine for all.

Admixture illuminates precision medicine inequities

Continental and ethnic classifiers are flawed proxies for ancestral diversity. In an increasingly interconnected world, rates of admixed ancestry (individuals of multiple ancestries) are likely to increase. Even in the present day, the extent and impact of admixture within human populations remains under-recognized, especially as it pertains to underrepresented groups in the United States. Moreover, our results show that, across all cancers, admixed individuals are less consistently classified than individuals assigned to a single ancestry group (one-sided unpaired t-test PhyloFrame model classification of admixed vs single ancestry individuals ($df = 109.93$, $p = 0.182$, effect size = 0.1, 95% CI = $-Inf - 2.427$); one-sided unpaired t-test benchmark model classification of admixed vs single ancestry individuals ($df = 110.49$, $p = 0.088$; effect size = 0.147, 95% CI = $-Inf - 0.987$); Supplemental Figure 8). Thus we sought to quantify and understand how admixture impacts the predictive power of PhyloFrame relative to the benchmark. We examined predictive efficacy of models trained on single ancestry data when applied to individuals of admixed ancestry. We explored the impact of admixture on model performance in the TCGA BRCA dataset, limiting our analysis to African, East Asian, and European ancestries (AFR, EAS, EUR and ADMIXED individuals with majority AFR, EAS, or EUR ancestry) due to the insufficient number of admixed individuals of other ancestries in the dataset. We selected breast cancer because the overall high AUC across all models allows us to largely exclude effects generated by poor model performance, and instead identify which individuals the models' struggle to accurately predict. Our analysis is restricted to models trained on a single ancestry group to precisely determine whether an individual is being stochastically or systematically incorrectly predicted. We applied all of the trained PhyloFrame and benchmark BRCA models to the 629 TCGA BRCA individuals with majority EUR ancestry, 122 individuals with majority AFR ancestry, and 38 individuals with majority EAS ancestry. Figure 4 shows individual level performance across all the models, sorted by percent majority ancestry of that individual (AFR, EUR, or EAS). Lines show the fraction of the models that correctly predict each majority AFR individuals' BRCA subtype, for PhyloFrame and benchmark. For each admixed individual, we calculated the fraction of models trained on EUR data that correctly predicted the individual's subtype and plotted this performance relative to their admixed ancestry proportion. We calculated statistical variance of model prediction accuracy across populations, using a LOESS smoothing model.

Overwhelmingly European training data sets may appear to perform acceptably in African ancestry individuals, when in fact performance is not uniformly high across the group. Although overall AUCs for PhyloFrame and benchmark models are very high (PhyloFrame mean AUC = 0.99993, Benchmark mean AUC = 0.99982) for all models and all ancestry test groups (Fig. 2), model recall is significantly higher in PhyloFrame compared to benchmark (Bonferroni adjusted one-sided paired t-test of BRCA model recall, PF vs BM ($df = 419$, $p = 2.464e-45$, effect size = 0.787, 99% CI = $0.118 - Inf$)). Sample-specific model accuracy is high and stable across admixed individuals with majority European ancestry (Fig. 4A), however, among individuals of majority

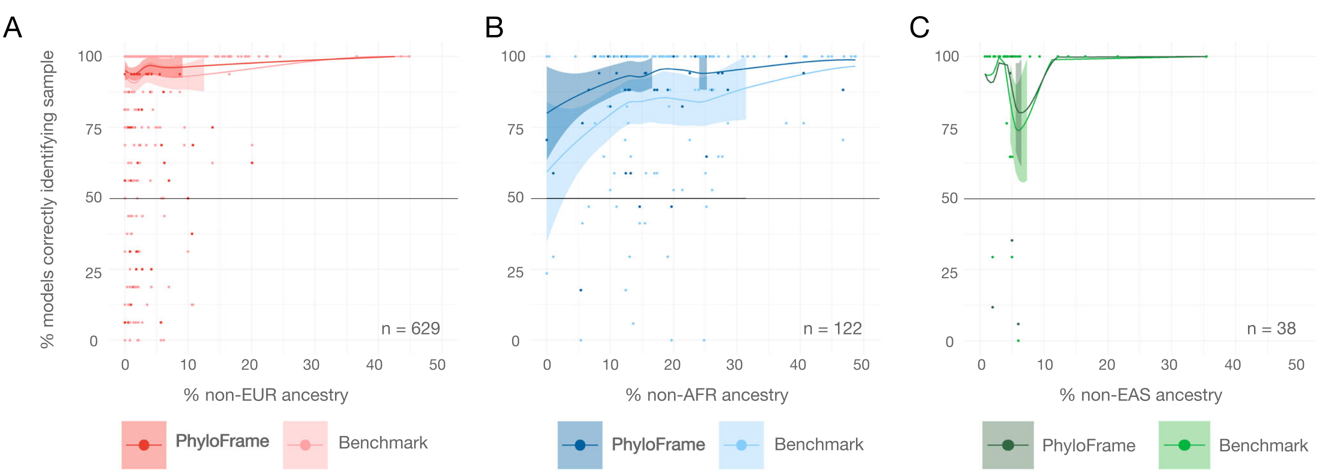

Fig. 4 | Admixture illuminates disparities in model performance. The fraction of all models correctly predicting breast cancer subtype in each patient with (**A**) majority European ancestry (629 EUR individuals), (**B**) majority African ancestry (122 AFR individuals), and (**C**) majority East Asian ancestry (38 EAS individuals). Individuals sorted by (x-axis) percentage of admixed ancestry. Smoothing with local polynomial regression fitting (LOESS) is used to fit a line to PhyloFrame and benchmark predictions in each majority ancestry, shown as solid dark lines in each plot. A 95% confidence interval for PhyloFrame and benchmark models is shown as a shaded bar surrounding each line. Source data are provided as a Source Data file.

African ancestry (Fig. 4B), PhyloFrame models have significantly higher sample-specific accuracy than benchmark models across ancestry levels. This suggests that the benchmark models are overfitting and modeling variations more prevalent in individuals of EUR ancestry rather than variations specific to breast cancer but seen in all ancestral populations. Of note, the performance of both PhyloFrame and the benchmark increase significantly as the percentage of European ancestry increases in majority African ancestry individuals, but PhyloFrame performs much better than the benchmark for individuals with very little or no European ancestry (Fig. 4B). We hypothesize that this improvement in model performance with increased admixed ancestry is a product of ancestral bias in the training data. The preponderance of admixed ancestry in African American populations is European in origin[52] (Supplemental Figure 9B). As such, performance of the models trained on EUR data improves as the individuals' fraction of European ancestry increases. This raises substantial concerns surrounding existing models' abilities to provide insightful and accurate precision medicine predictions for individuals of un-admixed African ancestry.

External validation of BRCA models

To validate model performance, we curated data from 14 unique breast cancer studies, totaling 1039 individuals of diverse ancestries (Table 1). Metadata provided by these studies was used to identify race or ethnicity information for each sample, with which we assigned continental ancestry. Where available, we also include more specific race and ethnicity information. The data was downloaded, re-processed, and combined using field standard approaches (see Methods). Using this combined data, we then applied each of the trained PhyloFrame and benchmark models to predict basal versus luminal subtype. Each of the 21 PhyloFrame and 21 benchmark BRCA models were applied to all 1039 samples. Figure 5 shows the per-individual performance across all models, calculated as the percentage of models that correctly identify each individual's subtype (sample-specific accuracy). Samples are grouped by continental ancestry (see Supplemental Figure 10 for performance differences grouped by race/ethnicity). PhyloFrame outperforms the benchmark when comparing all samples in this validation set (one-sided paired t-test of prediction accuracy, PF vs BM

Table 1 | External Validation Datasets

Study	Publication	Basal	Luminal	African	American	Asian	European	Other
BRCA_SMC_2018	Kan et al.[79]	36	112	-	-	148	-	-
GSE101927	Serrano-Gómez et al.[76]	-	42	-	42	-	-	-
GSE142258	Saleh et al.[55]	14	-	15	-	-	-	-
GSE142731	Saleh et al.[55]	42	-	23	-	-	19	-
GSE2109	-	50	184	11	10	2	208	3
GSE211167	Martini et al.[27]	26	-	26	-	-	-	-
GSE225846	Tang et al.[56]	15	62	52	-	1	24	-
GSE33095	Chen et al.[106]	9	53	-	-	62	-	-
GSE37751	Terunuma et al.[57]	14	7	-	-	7	-	-
GSE46581	Lindner et al.[58]	90	-	43	8	-	25	14
GSE59590	Huang et al.[107]	25	108	-	-	54	79	-
GSE62502	Vaca et al.[108]	12	-	-	12	-	-	-
GSE75678	Tamez et al.[109]	14	39	-	53	-	-	-
GSE86374	Romero et al.[74]	14	77	-	91	-	-	-

Dataset IDs, original publication associated with the study, the number of basal and luminal samples, and sample counts per continental ancestry group in each study that met our criteria for inclusion in the validation analysis. A hyphen (-) indicates no data.

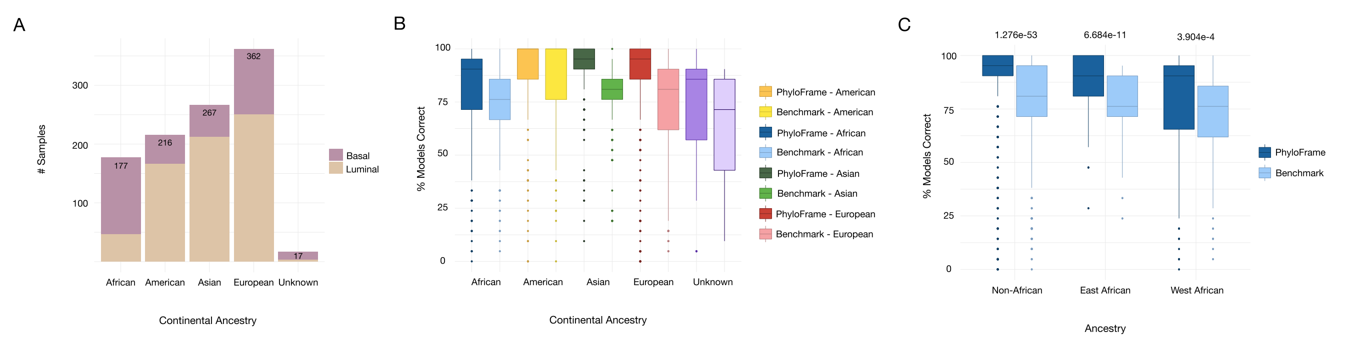

Fig. 5 | Validation in a global model. To assess sample level performance in the validation data, we show the percentage of models that correctly predict each sample. **A** Distribution of samples from each continental ancestry group and BRCA subtype in samples from the BRCA validation datasets. **B** Boxplots showing per-sample performance of the PhyloFrame and benchmark models, grouped by continental ancestry (177 African individuals, 216 American individuals, 267 Asian individuals, 362 European individuals, 17 Unknown individuals). The median is used as the measure of center for each box. **C** Boxplots showing performance in East African (Kenyan, Ethiopian), West African (Ghanaian, African American), and non-African samples (862 non-African individuals, 49 East African individuals, 128 West African individuals). Statistical significance of PhyloFrame versus benchmark performance within each ancestry group is shown above each boxplot. Statistical significance of PhyloFrame performance across the ancestry groups: one-sided unpaired t-tests (East African vs West African ($df = 146.23$, $p = 8.086e-04$, effect size = 0.477, 95% CI = 5.347 − *Inf*), non-African vs East African($df = 59.953$, $p = 0.486$, effect size = 0.004, 95% CI = − 4.081 − *Inf*), non-African vs West African ($df = 154.98$, $p = 1.924e-05$, effect size = 0.426, 95% CI = 6.777 − *Inf*)). Adjustments for multiple comparisons were not done. For all boxplots in (**B**) and (**C**), the median is used as the measure of center for each box with a minima of 0 and a maxima of 100. Lower and upper box bounds show the first (25th percentile) and third (75th percentile) quantiles. The top whisker extends to the largest value that is no larger than 1.5*IQR (inter-quantile range) from the top box bound and the lower whisker extends to the smallest value at most 1.5*IQR of the lower box bound. Points beyond whiskers are outliers. Source data are provided as a Source Data file.

($df = 1038$, $p = 1.524e-60$, effect size = 0.543, 95% CI = 7.543 − *Inf*)), and in all within-continental-ancestry-group comparisons (one-sided paired t-tests of prediction accuracy within each continental ancestry PF vs BM: African, 177 samples, (df = 176, p = 7.149e − 09, effect size = 0.447, 95% CI = 4.254 − *Inf*); American, 216 samples, (df = 215, p = 0.097, effect size = 0.089, 95% CI = − 0.261 − *Inf*); Asian, 267 samples, (df = 266, p = 2.657e − 38, effect size = 0.931, 95% CI = 11.717 − *Inf*); European, 362 samples, (df = 361, p = 1.346e − 26, effect size = 0.605, 95% CI = 8.981 − *Inf*); Unknown, 17 samples, (df = 16, p = 0.119, effect size = 0.297, 95% CI = − 2.376 − *Inf*)). There is also statistically significant difference in performance between PhyloFrame and the benchmark models when comparing between race/ethnicity information (Supplemental Figure 10). Together, this demonstrates that PhyloFrame leads to better prediction for individuals of all ancestries.

Diversity within Africa impacts model performance

The ongoing lack of diverse African data has limited the ability of precision medicine studies to accurately represent African populations. This unintended bias is a source of severe and often hidden predictive deficits, as African populations possess greater diversity than the entire rest of the world combined and, critically, are not a monophyletic group[12]. In fact, East African populations are more closely related to non-African populations than they are to West African populations, although they are substantially different from both groups[12]. However, a lack of diverse African data has severely limited the ability of precision medicine studies to reliably address these representation deficits. The majority of (already underrepresented) African ancestry individuals in genomic datasets are members of the African diaspora[1,52]. The sizable majority of enslaved people taken to the United States during the transatlantic slave trade were taken from Atlantic Africa[52–54] (Supplemental Figure 9). Consequently, East African individuals are generally less well represented in omics datasets collected in Europe and North America[37,52] and, when represented, are classified as falling within the African group despite significant genomic divergence from the better represented African populations.

Our validation dataset includes 49 East African individuals with breast cancer (38 Kenyans[55,56] and 11 Ethiopians[27]), allowing us to directly interrogate how divergence from common ancestry groups impacts performance of PhyloFrame and the benchmark in this severely undeserved group. For purposes of comparison we will consider individuals currently living in West Africa (6 Ghanaians[27] and the 122 African Americans[27,55–58]) to be of West African descent. We assessed the sample-specific predictive accuracy (the percent of models correctly predicting each sample) of PhyloFrame for three groups: non-African, East African, and West African ancestry. In both PhyloFrame and the benchmark, predictive accuracy was greatest in non-African ancestry individuals, intermediate in East African ancestry individuals, and lowest in West African ancestry individuals, following genetic divergence from the most represented populations rather than presence in the training data (Fig. 5C). PhyloFrame outperforms the benchmark in all groups (one-sided paired t-tests, PF vs BM; non-African (df = 861, p = 1.276e − 53, *effect size* = 0.562, 95% CI = 7.946 − *Inf*); East African (df = 48, p = 6.684e − 11, *effect size* = 1.16, 95% CI = 7.871 − *Inf*); West African (df = 127, p = 3.904e − 4, effect size = 0.304, 95% CI = 2.258 − *Inf*)) but the drop-off in performance in between PhyloFrame and the benchmark is greater in African populations (median PhyloFrame 90.5, benchmark 76.2; Fig. 5C).

Discussion

This study demonstrates that equitable machine learning techniques can insulate omics precision medicine models from ancestral bias, making them more robust when training on ancestrally homogeneous data. Our equitable machine learning framework, PhyloFrame, is better able to compensate for data distribution shifts caused by human ancestry differences in omics data. We demonstrate this in three TCGA cancer datasets and validate using data from fourteen other studies. In all three cancers, PhyloFrame has higher performance across several metrics. This holds true when considering the difficulty of prediction tasks, the degree of ancestry bias in the outcome labels, and the ancestry bias in the training data. PhyloFrame signatures are more consistent across models, demonstrating more stability and therefore likely more biological relevance than the comparable benchmark. Hard-to-predict samples are better served by PhyloFrame models than the benchmark. Furthermore, when considering individuals from ancestries that are severely underrepresented in the data (e.g. East African), PhyloFrame models are better able to accurately predict outcomes. The PhyloFrame framework is not in any way tailored toward these specific cancers or to cancer in general. It can generalize to many prediction tasks and diseases. When applying PhyloFrame to other diseases, we expect the largest performance improvements to be in diseases and outcomes impacted by ancestry bias. For example, PhyloFrame is unlikely to be informative for tasks with strong environmental factors, such as prediction of smoker vs non-smoker. With this in mind, Mendelian diseases are a good starting point for PhyloFrame application. Additional filtering, for example via analysis of EAF in disease-associated genes, can help identify diseases which may benefit the most from PhyloFrame. Selecting a prediction task with minimal label noise is also key for maximizing PhyloFrame's effectiveness. As shown in our results, subtype predictions in BRCA and UCEC, diseases with ancestry effects, perform better than the thyroid likelihood of metastasis. This is not to say that thyroid cancer does not have an ancestral effect, but does show that a difficult prediction task, with likely label noise, inhibits PhyloFrame's ability to target informative ancestry genes. We anticipate that PhyloFrame can play an important role in identifying which diseases and outcomes are most impacted by ancestry bias, and create equitable signatures of those outcomes and diseases.

Growing recognition of the impacts of ancestral diversity in human disease, the imbalance in sequenced populations and how this negatively affects precision medicine and other omics models have driven efforts to diversify sequencing[59–61]. The monumental scope of that task should not be understated. Artificial intelligence (AI) based precision medicine approaches require large number of individuals for training data and human diversity is not trivial to fully ascertain. Individual disease studies such as clinical trials will almost never be able to capture the full spectrum of human ancestral diversity in a single study. Given machine learning modeling relies on large-scale data, it is critical that sequencing efforts rise to this challenge and provide sufficiently diverse data to create equitable precision medicine[60,62–67]. We demonstrate that equitable machine learning is a powerful tool for mitigating biases in smaller studies. We also demonstrate the providing diverse information and using fairness machine learning techniques, amongst other approaches, to reduce ancestral bias in precision medicine modeling can result in improved outcomes for all, even individuals whose ancestry is over-represented in sequencing studies. We hope that these results will propel the field towards incorporating fairness techniques in omics models to mitigate these biases.

Overfitting to over-represented populations' standing genetic variation is one of the critical problems in precision medicine that we address with PhyloFrame. Our EAF statistic, calculated from the genetic variation of healthy individuals from across the globe, allows us to identify genes with elevated population-specific variation and amplify their likelihood of being included in the analysis. The effect, as we observe through PhyloFrame's improved performance relative to a comparable benchmark, is to force the models away from overfitting on genetic variation specific to populations over-represented in the disease-specific training data (usually European). Importantly, EAF does not specifically enrich for known cancer-related genes. This provides confidence that the benefits derived in PhyloFrame are not the result of simply capturing disease-related genes. Therefore, EAF should generalize well to other disorders. The EAF statistic is critical to integration of functional networks, which have been repeatedly shown effective for omics precision medicine modeling[68–70]. We show that, in PhyloFrame, the functional interaction network is critical for identifying which ancestry-enriched genes interact with the original disease signatures. PhyloFrame uses the functional interaction network to find communities of genes that are variable across human populations and also related to the outcome of interest to the model, such as cancer subtype. While we use the HumanBase tissue-specific functional interaction networks, there is potential to increase model performance by either using different networks or multi-layer network analysis to integrate not only ancestry enrichment, but also ancestry enrichment and its impacts across many tissues. Additionally, while we did not statistically evaluate whether or not EAF is enriched in other disease signatures, we believe that this would be of interest going forward as it is potentially a way to identify which diseases are most heavily impacted by ancestry bias or are truly enriched in specific ancestry groups. Some diseases are more likely to occur in specific ancestries, and there is the potential to use the EAF statistic to identify which genes or variants associated with the disease are ancestry-related and which are not. EAF provides additional statistical power for these types of analyses. The EAF statistic can be calculated using more or less granular ancestry definitions, allowing its use in pinpointing disease focal points in small populations.

We tested PhyloFrame against a comparable benchmark in three TCGA cancers. These were selected based on diversity of each TCGA cancer and to provide a range of difficulty in prediction tasks and the likely degree to which ancestry affects the predicted outcome labels. Label noise driven by ancestry bias in outcome label definition can have negative effects on model performance and cause negative data cascades[49,71], and so we sought to test PhyloFrame's ability to mitigate these challenges. The hardest prediction task, metastasis in thyroid tumors, suffers from low performance in both PhyloFrame and benchmark models, although PhyloFrame has slightly higher model performance. PhyloFrame's greatest impacts, however, are seen in the uterine and breast cancer analyses. In both BRCA and UCEC, PhyloFrame has higher performance and is better at individual-level predictions, even for samples that are from mixed ancestry or under-represented ancestry groups. Model performance shows significant improvement when using PhyloFrame models in both cancers, regardless of the ancestry of the train or test data. One easy prediction task, breast cancer subtype, is known to be unevenly distributed across ancestry groups and also to be critical to treatment options and outcomes. Basal breast cancers are more likely to arise in people of African descent than others, however it presents in individuals from all ancestries. The basal subtype is harder to treat and has worse outcomes[27]. When analyzing PhyloFrame performance in predicting basal versus luminal breast cancer subtype, we found that PhyloFrame has significantly higher performance across most metrics. Beyond cohort level performance, we also tested per-individual model performance and found that in all three cancers, there is a large variance in performance which correlates with ancestry; higher levels of over-represented ancestries results in higher likelihood of the models accurately predicting that individuals subtype, although this limitation was more stark in the benchmark models. Individual level performance in admixed individuals was also illuminating. In both the benchmark and PhyloFrame, model performance increased as the level of European ancestry increased. However, PhyloFrame was better able to counter these effects and performed better across all ancestry groups in all cancers.

Many recent calls to action have highlighted disparities in precision medicine model performance across ancestry groups. To assess this in PhyloFrame, we collated data from 14 studies, with a total of 1039 samples from around the world. We applied PhyloFrame and the benchmark models to this data to assess overall performance for each individual in the validation study. We identified statistically significant differences in performance in continental ancestry groups. Across all samples and within each ancestry group, PhyloFrame performance is statistically significantly higher than the benchmark. Our validation dataset also allows us to consider how populations that are poorly represented in most omics datasets perform in PhyloFrame and the benchmark. East African populations are genetically distinct from both West African and non-African populations but are overall more closely related to non-African than West African populations[12]. East African populations are also severely underrepresented in sequencing databases because most individuals of African ancestry in the United States and Europe are not of East African ancestry[52,53]. PhyloFrame and benchmark models are most accurate in non-African ancestries, then lower in East African ancestries and lowest in West African ancestries. This matches their relative genetic divergence rather than representation in sequencing databases. While both types of model are affected by this, PhyloFrame has higher performance than the benchmark in all groups, and PhyloFrame has a smaller drop in performance when considering African individuals.

We chose to delve into model performance in eastern versus western African ancestry because it provides a unique opportunity to test the efficacy of our models in severely underrepresented populations. Individuals with ancestries that are not well represented in wealthy countries, in particular, are at greater risk of receiving lower quality precision medicine care under non-ancestry-aware methods (e.g. the benchmark). Africa is the most severely impacted continent in this regard due to a combination of economic disadvantages and greater genetic diversity than the rest of the continents combined[12]. Better representation of African ancestry is a critical challenge across human genetic studies. Africa is home to the majority of human genetic diversity, with non-African diversity being only a subset of African genetic diversity[12]. However, the continent is severely undeserved in terms of genetic and biomedical research[72,73]. All analyses of human genetic disorders would likely be better served if there was sufficient data to capture variation within Africa. Dataset quality differences may account for some of the variance in the PhyloFrame and benchmark models' performance when applied to the African samples. However, PhyloFrame shows statistically significant improvement over the benchmark model when it comes to accurately predicting breast cancer subtypes within African data, the most diverse human continental population.

PhyloFrame is an important advance relative to prior attempts to address bias in precision medicine. Prior studies have advanced predictive efficacy by developing different disease signatures for each major ancestry group[41,74–81]. However, that approach poorly serves individuals with diverse ancestries or ancestries without sufficient data to train a model. Multi-racial individuals or individuals from severely under-sequenced regions, such as East Africa and Oceania, are unlikely to benefit from such an approach as those groups are rarely represented in large enough numbers to train a model in wealthy countries where most precision medicine research occurs. Additionally, these single ancestry models fail to leverage the information inherent to human genetic diversity, thus leaving a significant source of information untapped. PhyloFrame resolves this issue by developing a single disease signature for all individuals, regardless of ancestry, informed by global population diversity. This utilizes more data and information while also preventing user errors, as it is not necessary to manually assign ancestry to an individual in order to determine which single-ancestry model best applies to them. As we show in our validation analysis, PhyloFrame yields substantial benefits globally. This is true for both over-represented and under-represented populations, including those that were not represented in the training data (American, East African), and in admixed individuals.

Populations are a valuable framework for assessment of human diversity but they should not obscure the complexity of human history. Admixture between nearby populations is a constant phenomenon. An oversimplification in ancestry assignments specifically disadvantages individuals descended from more than one of the assigned populations. Our exploration of the impact of admixture demonstrates that PhyloFrame improves performance for individuals with complex ancestries. Our analysis of the impact of European admixture in African American TCGA breast cancer patients demonstrates that the exact distribution of ancestry can exert a significant impact on model performance on a given individual. PhyloFrame provides a substantial improvement in model equity and overall performance over the benchmark. While there is still room for improvement, PhyloFrame is an important step in moving toward precision medicine frameworks that are not dependent on prior knowledge of patient ancestry for optimal performance.

Unbalanced population diversity within accessible genomic data has led to unintended bias in precision medicine models[13,30,82–85], resulting in disparate effectiveness in populations and sub-optimal treatment options[16,18,86–88], and contributing to the inequality of medical resources[15,17,18,89–92]. PhyloFrame mitigates these issues through equitable machine learning, creating genomic signatures of disease that are equally effective in all populations regardless of the ancestry of available training data. PhyloFrame demonstrates the potential of its type; one that integrates population genomics data and functional interaction network modeling to improve disease prediction by eliminating overfitting toward over-sampled populations. Using an equitable machine learning technique and a well-benchmarked and commonly applied underlying machine learning method that is robust to data distribution shifts and small sample sizes, PhyloFrame is able to correct for ancestral bias in signatures, creating disease signatures that work better for all individuals. Its signatures are more consistent across training sets (demonstrating higher biological relevance) and consistently detect known cancer-related genes. Unlike a comparable benchmark, PhyloFrame is effective regardless of whether an individual comes from a population represented in the training data. Overall PhyloFrame performance for all populations is higher than the comparable benchmark. It performs well in individuals of divergent unsampled populations and varying levels of admixture. This not only brings performance in under-sampled populations to equality with better sampled populations but provides better performance to all populations, across the spectra of over- and under-representation. Integrating population genomic data with disease oncology presents a unique opportunity for novel approaches in the examination of fundamental mechanisms of disease.

PhyloFrame occupies a critical, and to this point unfulfilled, niche in the broader movement towards more equitable precision medicine. It demonstrates the feasibility of equitable genomic machine learning approaches and serves as a catalyst to improve precision medicine equity, alongside improved genomic references that incorporate population structure and diversity (such as the Human Pangenome Reference Consortium[93,94]) and sequencing efforts (such as the Nigerian 100K Genome Project[95]). Consequently this study does not capture the full range of potential applications and refinements of the method. For example, the use of network-based machine learning approaches and additional machine learning fairness techniques designed to correct for bias and misinformation[96–98], that have great potential in the genomics space. In addition, deep learning models have been shown to implicitly encode learning algorithms such as linear regression in their activation layers and to converge to Bayesian inference as the layers scale[99]. As such, the techniques used in PhyloFrame should translate to other machine learning approaches and the lessons learned here apply to many types of machine learning methods. Consequently, PhyloFrame demonstrates the benefits of enforcing population variability in during signature creation and represents a major step forward. Combined with continually improving genomic resources and data, it will further enable a future of equitable precision medicine, where all individuals can trust that models will accurately predict their genomic information. We anticipate that future studies will expand upon both of these areas yielding further improvements in performance over non-ancestry informed models across a wide range of precision medicine applications.

Methods

Population Genomics Compendium

Population level allele frequency (AF) is calculated using using sequence variation information from The Genome Aggregation Database (gnomAD)[100]. GnomAD compiles data from many extensive sequencing projects around the globe and is the largest reference population database currently available. The gnomAD v4.1 data used by PhyloFrame includes 730,947 exome sequences from diverse, unrelated individuals. PhyloFrame uses the following 8 ancestries defined by gnomAD, which follow NASEM[44] guidelines: African/African American (16,740 individuals), Ashkenazi Jewish (13,068 individuals), Admixed American (22,362 individuals), East Asian (19,850 individuals), European (Finnish) (26,710 individuals), European (non-Finnish) (556,006 individuals), Middle Eastern (2,884 individuals), and South Asian (43,129 individuals). We exclude all other individuals without a specific defined ancestry, annotated as 'Remaining' by gnomAD (30,198 individuals). GnomAD assigns ancestry by first using principle component analysis (PCA) to cluster genomically similar individuals. A random forest model is then trained on samples with known ancestry labels using the defined principle components from the PCA as features. This trained model is used to assign all other samples to one of the gnomAD v4 ancestry groups[101]. All analyses in this manuscript use the full gnomAD dataset, not one of the provided subsets (e.g. UK Biobank). The gnomAD v4.1 exome variant calling format (VCF) files for each chromosome were downloaded from the gnomAD download page in May, 2024. Variant information for each ancestry as well as the gene information for each Single Nucleotide Polymorphism (SNP) were parsed in C++ using the VCF operations library htslib[102] and written to a new VCF file. For each SNP, the new VCF contains the allele count, allele number, and allele frequency for each of the 8 ancestries as well as the chromosome, position, rs id, reference allele, alternate allele, gene, consequence, impact, and distance. GnomAD takes consequence annotations (the fields including consequence, impact, and distance) from Ensembl VEP[103], and are defined by Ensembl VEP as follows: consequence is the consequence type of this variant, impact is the impact modifier for the consequence type, and distance is the shortest distance from variant to transcript. Allele counts, numbers, and frequencies are not sex-specific within the ancestry. The full pipeline for parsing population data from gnomAD (from file download to enhanced allele frequency calculation) is available via a Snakefile in the PhyloFrame GitHub repository.

Enhanced Allele Frequency

To calculate ancestry enrichment for each SNP, we create a statistic called Enhanced Allele Frequency (EAF). EAF identifies SNPs in exons that vary between global populations, and is based on allele frequency (AF). In the current study, EAF is calculated using allele frequency data from gnomAD (see above section, Population Genomics Compendium). For each SNP we calculate EAF for each ancestry as follows: Given k ancestries:

$$\mathrm{EAF}_i = \mathrm{AF}_i - \frac{1}{k-1}\sum_{i \neq j}^{k} \mathrm{AF}_j \qquad i = 1, \ldots, k \tag{1}$$

where AF_i equals the allele frequency of ancestry i and AF_j equals the allele frequency of ancestry j.

EAF shows which SNPs are enhanced, i.e. occur more frequently, in which ancestries and helps PhyloFrame target ancestry-specific genes that may affect the way an ancestry responds to given a disease. High EAF indicates that the given variant is more frequently altered in that ancestry, compared to other ancestries. Density plots of the resulting enhanced frequencies in each ancestry demonstrate that while most SNPs are negatively enhanced, there are ancestry unique peaks of enhanced SNPs (Supplemental Figure 3). EAF thresholds used in PhyloFrame were selected based on the distribution of EAF SNPs across all coding genes. In order to target ancestry-specific variation, we restrict to SNPs that are uniquely enhanced in any given ancestry (EAF between 0.2 - 1; Supplemental Figure 4). Final gene inclusion in PhyloFrame signatures is determined by the functional interaction network walk, EAF sorting, and variability in the training data.

PhyloFrame equitable machine learning framework

The PhyloFrame pipeline for population data is implemented in C++ and R. Population variation data (gnomAD) and functional interaction networks (HumanBase) are internal to PhyloFrame and users do not need to provide or process these information. However, we also provide processing scripts to create the correct formats (see Code Availability). PhyloFrame's two required inputs are gene expression data for a set of samples and outcome labels with which to train the model. Using this information, PhyloFrame creates a disease signature, then uses the population genomics data and functional interaction networks to modify that disease signature to be equally effective across all global populations. PhyloFrame returns this equitable disease signature, including weights describing the model importance for each gene. This process is described in detail below.

Functional Interaction Network Integration. For each disease, HumanBase functional interaction networks[104] were used to find genes likely to interact with the identified disease-associated genes. After the disease relevant functional interaction network was downloaded, Entrez IDs were mapped to Gene Symbols in R using Bioconductors' genome wide annotation for Humans (org.Hs.eg.db, Bioconductor version 3.17). Genes from PhyloFrame's baseline model signature (see Fig. 1) were used as start nodes in the search for disease relevant gene interactions in the functional interaction network. First and second neighbors of the baseline signature with a connection between 0.2 − 0.5 are kept for further ancestry allele sorting. By projecting high profile disease associated genes into a functional interaction network, PhyloFrame is able to begin its ancestry search in genes with high involvement in the disease. For each of the three cancers in this manuscript, PhyloFrame used the most relevant functional interaction network for each disease: mammary epithelium for BRCA, thyroid gland for THCA, and uterine endometrium for UCEC. PhyloFrame functional interaction network edge cutoffs were initially selected based on a grid search within each disease to find the optimal number of neighbors in the functional interaction network as well as the optimal edge weight for the interaction. PhyloFrame was run on every combination of the grid N × E where N = {1, 2, 3} defining any gene within three neighbors of a gene from PhyloFrame's baseline elastic net run and E = {0.5, 0.55, 0.6, 0.65, 0.7, 0.75, 0.8, 0.85, 0.9,.95, 1} defining the minimum edge weight needed for a neighbor to be included. PhyloFrame was run on this grid search for every ancestry in each disease, the combination with the highest overall average across all ancestral models was selected. Based on this search, we opted to included from the functional interaction network the first and second degree neighbors of the original signature genes. We later updated the edge weights to keep nodes with a weight between 0.2-0.5, based on analysis of the functional interaction between the European and African TCGA BRCA disease signatures.

EAF network walks. The EAF network walks work by first using a logistic regression model with LASSO penalty to find an initial signature of *k* disease-relevant genes. This results in a median signature size of 25 genes. For each gene in the original signature, PhyloFrame projects the original signature's genes onto the functional interaction network that has already been pruned based on edge weights. For each gene in the original signature, PhyloFrame extracts the entire list of genes that are up to two neighbors away from the gene of interest. Using this list of genes, PhyloFrame checks which have ancestry-specific EAF enriched SNPs. It creates one list of EAF enriched SNPs per ancestry within the set of identified neighbor genes. This process is repeated for all genes, and PhyloFrame then adds all neighbors to potential expanded signature set. For each ancestry list, SNPs are converted to gene IDs. For each ancestry gene set, PhyloFrame selects the top 30 most variable genes, using the training data expression values, and adds those 30 genes to final PhyloFrame signature. PhyloFrame selects the 30 most variable genes in each ancestry gene set in order to approximately match the size of the original LASSO gene signature and ensure each ancestry has equal representation in the final gene signature. Signature weights for all genes in the final signature (both original signatures genes and those added by PhyloFrame) are then recalculated. The final signature is run with ridge penalty for forced inclusion of ancestry genes. By including 30 most variable genes per ancestry, the model is forced to equally weight all ancestries. This partially mitigates over-representation of some ancestries in the gnomAD data. Increasing or decreasing the number of variable genes included in the original signature set does not have a large impact on the performance of PhyloFrame. Decreasing the number of variable genes results in a smaller final signature size and includes less ancestry-specific genes, restricting the scope of ancestry-relevant genes to investigate. Significant reduction in the number of variable genes included will cause the model to primarily rely on the initial LASSO signature. Increasing the number of variable genes has the opposite effect, creating a larger signature size with more ancestry-relevant genes. However, as all genes are forced to be included in the final signature, large increases in the number of variable genes included comes with the risk of forcing inclusion of less disease relevant genes into the final gene signature. Figure 1 visualizes the approach.

Integrating ancestry and functional interaction networks. For any disease, PhyloFrame requires the ancestry-specific EAFs for all genes, the tissue associated functional interaction network, patient gene expression data, and patient clinical data. Using the EAF information and a functional interaction network, PhyloFrame first finds the neighborhood of the top enhanced genes by EAF, using node and edge parameters previously found in each diseases' grid search. The resulting neighborhood of genes are mapped to their associated SNPs using the previously calculated EAFs. We narrow down the disease's genomic landscape further by ordering the enhanced allele frequencies for each ancestry separately and targeting the unique allele frequency in each ancestry with the most enhanced SNPs. The genes included in this slice for each ancestry are assembled and prioritized in PhyloFrame's regression task. To do this, PhyloFrame first narrows the genomic landscape by selecting genes enhanced in each ancestry for genes relevant to disease in each ancestry by looking at the enhanced allele frequency, it then uses logistic regression with a ridge penalty to select highly important genes. It returns to the user an equitable gene signature of disease.

Equitable loss function for machine learning. PhyloFrame builds ancestry unbiased disease signatures by minimizing loss in two domains: relevance to the given disease and enrichment in at least one of the 8 ancestries described by gnomAD. To define the disease-related genes, we find genes that are functional network interaction partners with the genes identified by PhyloFrame's baseline elastic net run. Supplemental Figure 2 shows the related signatures for breast cancer subtype prediction when training in two ancestral populations. To create the PhyloFrame signatures for this task, we created a signature of disease that does not incorporate ancestry, and instead simply calculates

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relationship within the training data samples to the disease outcomes. We use these genes as start nodes in the search for disease-relevant gene interactions in the functional interaction network. PhyloFrame performs a grid search to find fully connected subnetworks within the undirected functional interaction network graph. Genes are selected if they are within y neighbors of the disease genes, narrowing the genomic landscape by selecting genes enhanced in each ancestry for genes relevant to the disease by looking at the enhanced allele frequency, then logistic regression with a ridge penalty to select critical genes.

The benchmark model

Each benchmark model is trained similarly to PhyloFrame models, and this process is visualized in Supplemental Figure 5. Benchmark models use the same initial gene signature as PhyloFrame. This signature is created based on the training data using a logistic regression model with LASSO penalty. The signature is then increased in size to match the corresponding PhyloFrame model's final signature size (after network propagation and EAF filtering) by adding genes that are highly variable in the training data. The final benchmark signature is a combination of the original signature genes (the same genes as the original signature in the PhyloFrame model) and highly variable genes. The benchmark model is then retrained and forces the inclusion of this set of genes using a ridge regression penalty. As a result, benchmark models have exactly the same signature size as PhyloFrame models trained on the same data, and have at least some signature overlap.

Prediction tasks and data setup for model training and testing

In order to evaluate PhyloFrame and benchmark model performance within and across ancestral populations, samples from each of the three cancers were divided into smaller independent, single ancestry datasets. All models were trained on one dataset and applied separately to each other dataset separately, resulting in 2802 sets of performance metrics (1401 PhyloFrame and 1401 benchmark). To do this, samples were divided into continental ancestry groups (European, EUR; African, AFR; East Asian, EAS; or Admixed, ADMIX) as assigned by Carrot-Zhang et al.[30] using their computational ensemble ancestry-calling prediction method, which adheres to NASEM guidelines. The sample size of the continental group with the smallest number of samples was used to determine the size of all independent single-ancestry datasets. Samples were randomly assigned to datasets, and each sample is included in one and only one dataset. Dependent variables were distributed as equally as possible across datasets. Datasets for each cancer are described below:

Subtype prediction in breast cancer (BRCA). Cancer subtype prediction is a key component of directing the treatment of breast cancer. We trained 21 models on data from 842 individuals, divided into training batches of 38-48 samples each, to predict basal vs luminal subtype. The number of training sets correspond to the proportion of samples in the TCGA BRCA database resulting in 17 EUR, 2 AFR, 1 EAS, and 1 ADMIXED models. PhyloFrame uses the HumanBase mammary epithelium network.

Subtype prediction in uterine cancer (UCEC). Cancer subtype prediction is a key component of directing the treatment of uterine cancer. We trained 16 models on data from 491 individuals, divided into training batches of 14-18 samples each, to predict endometrioid vs serous subtype. The number of training sets correspond to the proportion of samples in the TCGA UCEC database resulting in 12 EUR, 2 AFR, 1 EAS, and 1 ADMIXED models. PhyloFrame uses the HumanBase uterine endometrium network.

Metastasis prediction in thyroid cancer (THCA). Metastases have a substantial impact on patient prognosis, in this analysis we predicted whether patients would undergo metastasis (M0 versus MX). We trained 28 models on data from 436 individuals, divided into training batches of 14-18 samples each. The number of training sets correspond to the proportion of samples in the TCGA THCA database resulting in: 23 EUR, 1 AFR, 3 EAS, and 1 ADMIXED models. PhyloFrame uses the HumanBase thyroid gland network.

External validation of BRCA models. To externally validate our models, we compiled data from ancestrally diverse breast cancer studies (listed in Table 1). This provided an opportunity to assess the performance of PhyloFrame and the benchmark on ancestry groups not present in the training data. We searched cBioPortal[105], dbGAP, and the Gene Expression Omnibus (GEO) for studies with ancestry and subtype information. Fourteen studies met that criteria, totaling 1,039 samples. Metadata for all studies was compiled manually to find attributes common among all studies. Where possible, samples with subtype information (such as hormone receptor status) but without PAM50 subtype were assigned a basal or luminal subtype. Additionally, we assigned a continental level ancestry based on race/ethnicity data provided by the original study. Original subtype and ancestry information for each sample along with our assigned values can be found in Supplementary Data 3 Each study was mapped to their corresponding platform IDs to obtain gene information when necessary followed by gene name conversion to Gene Symbols using the Bioconductors' genome wide annotation for Humans (org.Hs.eg.db, Bioconductor version 3.17) in R. After each dataset had Gene Symbols, all datasets were merged in R using Gene Symbols. As with the previous analyses of breast cancer, PhyloFrame and the benchmark were tasked with classifying the samples as either basal or luminal. The final expression matrix used as input to PhyloFrame and the benchmark is in Supplementary Data 4. Due to minor variations in genes included in the validation studies, we retrained the PhyloFrame and benchmark models trained on the independent TCGA datasets defined above (see section Evaluating model performance across many independent datasets). We applied the PhyloFrame and benchmark models trained using the same subdivisions of the TCGA BRCA data described above, resulting in 21 models (17 EUR, 2 AFR, 1 EAS, and 1 ADMIXED). These trained models were applied to the external validation set.

Statistical Analysis

All statistical tests were calculated in R. A paired Student's t-test was used for paired comparisons, and an unpaired t-test was used for comparisons between two independent groups. A two-sided unpaired t-test was used to compare EAF enrichment between COSMIC genes and non-COSMIC genes. One-sided unpaired t-tests were used in all comparisons of sample classification for individuals of admixed ancestry vs those assigned to a single ancestry. A one-sided paired t-test was used in comparisons of model performance metrics Matthew's Correlation Coefficient (MCC), Area under the ROC curve (AUC), F-measure, Cohen's kappa, recall, and accuracy. A one-sided paired t-test was used for all sample classification comparisons in the validation dataset. All computational experiment comparisons include statistical test type, degrees of freedom, p-value, effect size, and confidence interval and are reported in the corresponding text and figures. The threshold for significance was set at $p < 0.05$ after Bonferroni correction for multiple testing. Plots are generated in R using ggplot2. For boxplots, each boxplot marks the first and third quartiles with a line in the colored bar marking the median value. Error bars show Q1/Q3 *1.5 the interquartile range, respectively. Outliers are shown as independent points. Smoothed line plots are shown with Loess smoothing with a 95% confidence interval. Smoothed line plots are shown with Linear smoothing with a 95% confidence interval for UCEC recall.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All analyses for this study were conducted on published datasets. Population genomics data from gnomAD (gnomad.broadinstitute.org) was downloaded in May 2024. We downloaded functional interaction networks for model generation from HumanBase (hb.flatironinstitute.org); the mammary network was downloaded in July 2022 and uterine and thyroid networks were downloaded in December 2022. Predictive model training (PhyloFrame and benchmark) and all analyses except the validation experiment (Fig. 5) use gene expression data and matching clinical data from the TCGA Pancancer Atlas for breast (BRCA), thyroid (THCA) and uterine (UCEC) cancers. Data for all TCGA samples was accessed from cBioPortal (www.cbioportal.org) in December 2022. Ancestry information for the TCGA cancer data was obtained from Carrot-Zhang et al.[30]. Datasets used in the BRCA external validation (Table 1) were downloaded from cBioPortal and GEO in April 2024. The complete gene expression matrix for the validation set used in this study is provided in the Supplementary Information. Accession numbers and links to each individual dataset used in the validation set are listed here: BRCA_SMC_2018, GSE101927, GSE142258, GSE142731, GSE2109, GSE211167, GSE225846, GSE33095, GSE37751, GSE46581, GSE59590, GSE62502, GSE75678, GSE86374. COSMIC data was downloaded in January 2023 from the COSMIC website (cancer.sanger.ac.uk/cosmic). It contains 734 genes and information on which cancers each gene has been associated with. The enhanced allele frequencies used in this analysis, calculated from GnomADv4.1 are available for download as a tsv file on Zenodo under https://doi.org/10.5281/zenodo.14180045. Source data are provided with this paper.

Code availability

PhyloFrame is distributed under GNU General Public License v3.0. All code for the PhyloFrame equitable machine learning model and comparable benchmark model can be accessed at github.com/lesliesmith1112/PhyloFrame as well as on Zenodo under https://doi.org/10.5281/zenodo.14034516. Directions for installing the PhyloFrame R package are included in the GitHub repository.

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Author contributions
K.G., L.A.S., and J.A.C. designed the study, conceptualized the method, and planned experiments. L.A.S. and K.G. developed the software and conducted experiments. K.G., L.A.S., and J.A.C. analyzed and interpreted results. JL provided statistical support. K.G., J.L., and J.A.C. acquired funding and supervised the study. K.G., L.A.S., and J.A.C. wrote the manuscript and visualized data. All authors approved the final manuscript.

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The authors declare no competing interests.

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